DAY 2 - BEGINNER

SUM, AVERAGE, MEDIAN, MIN, MAX & COUNT

Primary tool: Google Sheets

What you are learning

Basic descriptive statistics and the difference between average and median.

Learning objectives

- Write basic formulas

- Choose the right summary statistic

- Check whether a result is reasonable

Worked practical

Calculate score summaries from a participant test table and change source values to confirm formulas update.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Ask AI: 'Explain why mean and median may differ. Give me a new example, but do not solve my dataset.'

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Summarize test scores and explain mean vs median.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.